1. Solid State

Solids are materials with **definite shape and volume**. Particles are **closely packed** and vibrate about fixed positions. Solids are classified as **crystalline** or **amorphous** based on the arrangement of particles.

1.1 Amorphous & Crystalline Solids

- **Crystalline solids:** Particles are arranged in a **long-range, regular, repeating pattern**.
  - Examples: NaCl, Diamond, Quartz.
  - Properties: Sharp melting point, anisotropic (properties vary with direction).
- **Amorphous solids:** Particles are **randomly arranged** without long-range order.
  - Examples: Glass, Rubber, Plastic.
  - Properties: Gradual melting, isotropic (same properties in all directions).

1.2 Crystal Lattices & Unit Cells

- **Crystal lattice:** 3D arrangement of atoms, ions, or molecules in a crystal.
- **Unit cell: Smallest repeating unit** of a crystal lattice that defines its structure.
- **Types of unit cells:**
  1. Simple cubic (SC)
  2. Body-centered cubic (BCC)
  3. Face-centered cubic (FCC) / Cubic close-packed (CCP)

1.3 Number of Atoms in Unit Cell

- **SC:** 1 atom (8 corners × 1/8)
- **BCC:** 2 atoms (8 corners × 1/8 + 1 center)
- **FCC:** 4 atoms (8 corners × 1/8 + 6 faces × 1/2)

1.4 Packing Efficiency

- Fraction of space occupied by atoms in a unit cell.
- Formula:

$$\text{Packing efficiency} = \frac{\text{Volume of atoms in unit cell}}{\text{Volume of unit cell}} \times 100$$

- **SC:** 52.4%
- **BCC:** 68%
- **FCC:** 74%

1.5 Voids (Tetrahedral & Octahedral)

- **Tetrahedral void:** Formed when **4 atoms are in contact**; 2 atoms can fit in 8 tetrahedral voids.
- **Octahedral void:** Formed when **6 atoms surround a void**; 1 atom can fit in 4 octahedral voids.
- Voids are important in **ionic solids** for locating smaller ions.

1.6 Density of Unit Cell

$$\text{Density} = \frac{\text{Mass of atoms in unit cell}}{\text{Volume of unit cell}}$$

$$\rho = \frac{n \cdot A}{V_c \cdot N_A}$$

Where:

- n = number of atoms per unit cell
- A = atomic mass
- V_c = volume of unit cell
- N_A = Avogadro's number

1.7 Imperfections in Solids

- **Point defects:** Vacancy, interstitial, substitutional.
- **Line defects:** Dislocations (edge, screw).
- **Surface defects:** Grain boundaries, stacking faults.
- **Effects:** Can alter **density, conductivity, strength, color, etc.**

1.8 Electrical & Magnetic Properties

- **Electrical:** Depends on electron mobility.
  - Conductors, semiconductors, insulators.
- **Magnetic:** Based on electron spin and alignment.
  - Ferromagnetic, paramagnetic, diamagnetic, antiferromagnetic.

If you want, I can also make a **super short “1-page formula + key points sheet”** for all of these subtopics for **ultra-fast revision**. It would be perfect before exams.

Do you want me to do that?